

FYS5419/9419, lecture April 7

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$$\begin{aligned}
 & GFT |x_0 x_1 x_2 \dots x_{n-1}\rangle = \\
 & \frac{1}{\sqrt{2^n}} \left(|0\rangle + e^{2\pi i \phi_0 x_0} |1\rangle \right) \otimes \\
 & \left(|0\rangle + e^{2\pi i \phi_1 x_1} |1\rangle \right) \otimes \\
 & \quad \vdots \\
 & \otimes \left(|0\rangle + e^{2\pi i \phi_{n-1} x_{n-1}} |1\rangle \right)
 \end{aligned}$$

$n = 2$ example

$$|00\rangle = |0\rangle \otimes |0\rangle = |0\rangle_2 \quad (|0\rangle)$$

$$|01\rangle = |0\rangle \otimes |1\rangle = |1\rangle_2$$

$$|10\rangle = |1\rangle \otimes |0\rangle = |2\rangle_2$$

$$|11\rangle = |1\rangle \otimes |1\rangle = |3\rangle_2$$

$$|x\rangle_2 = |x_0 x_1\rangle \quad \wedge \quad |y\rangle_2 = |y_0 y_1\rangle$$

$$|x\rangle_2 = \frac{1}{2} \sum_{y=0}^3 \exp \left[2\pi i' x y / 2^2 \right] |y\rangle_2$$

$$|x\rangle_2 = |x_0 x_1\rangle \quad x = 2 \cdot x_0 + x_1$$

$$10>_2 : \overset{x_0}{2} \cdot \overset{x_1}{0} + 0 = 0$$

$$11>_2 : 0 \cdot 2 + 1 = \underline{1}$$

$$12>_2 : 1 \cdot 2 + 0 = 2$$

$$13>_2 : 1 \cdot 2 + 1 = 3$$

$$\exp \left[\underbrace{2\pi i' x y / 2} \right]$$

$$= \frac{2\pi i'}{4} (2x_0 + x_1) (2y_0 + y_1)$$

$$= 2\pi i' \left(y_0 \left(x_0 + \frac{x_1}{2} \right) + y_1 \left(\frac{x_0}{2} + \frac{x_1}{4} \right) \right)$$

$$y_0, y_1, x_0, x_1 \in \{0, 1\}$$

note that

$$e^{2\pi i x_0 y_0} = \underline{1}$$

$$\left(\frac{1}{\sqrt{2}} \sum_{y_0=0}^1 \exp \left[2\pi i y_0 \frac{x_1}{2} \right] \right)$$

$$\times \left(\frac{1}{\sqrt{2}} \sum_{y_1=0}^1 \exp \left[2\pi i y_1 \left(\frac{x_0}{2} + \frac{y_1}{4} \right) \right] \right)$$

$$\times \underbrace{|y_0 y_1\rangle}$$

$$|y_0\rangle \otimes |y_1\rangle$$

1st term

$$\frac{1}{\sqrt{2}} \left(|0\rangle + e^{i\pi x_1} |1\rangle \right)$$

$$x_1 = 0 : \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$x_1 = 1 : \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

need to multiply with

$$\otimes \left(\frac{1}{\sqrt{2}} \sum_{y_1=0}^1 \exp \left[2\pi i y_1 \left(\frac{x_0}{2} + \frac{x_1}{4} \right) \right] \times |y_1\rangle \right)$$

This is a Hadamard operation on the first qubit

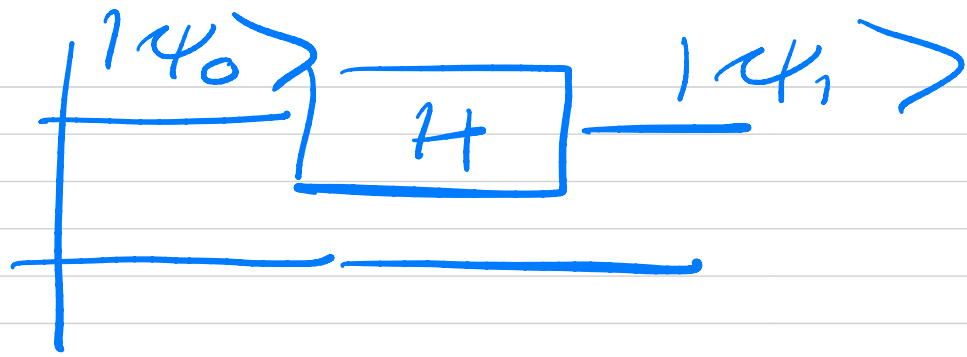
$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$H|0\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$H|1\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

First term is equivalent to

$$H \otimes I |y_0 y_1\rangle$$



$$|\psi_1\rangle = H|\psi_0\rangle$$

2nd term

$$\frac{1}{\sqrt{2}} \sum_{y_1=0}^1 \exp \left[2\pi i y_1 \left(\frac{x_0}{2} + \frac{x_1}{4} \right) \right]$$

$\times |y_1\rangle$

$$= \frac{1}{\sqrt{2}} \sum_{y_1=0}^1 \exp(\pi i y_1 x_0) \exp\left(\frac{\pi i y_1 x_1}{2}\right)$$

$\times |y_1\rangle$

Hadamard gate

Rotation gate

$$R_k = \begin{bmatrix} 1 & 0 \\ 0 & e^{2\pi i / 2^k} \end{bmatrix}$$

Here with $n=2$ $e^{\pi i / 2}$

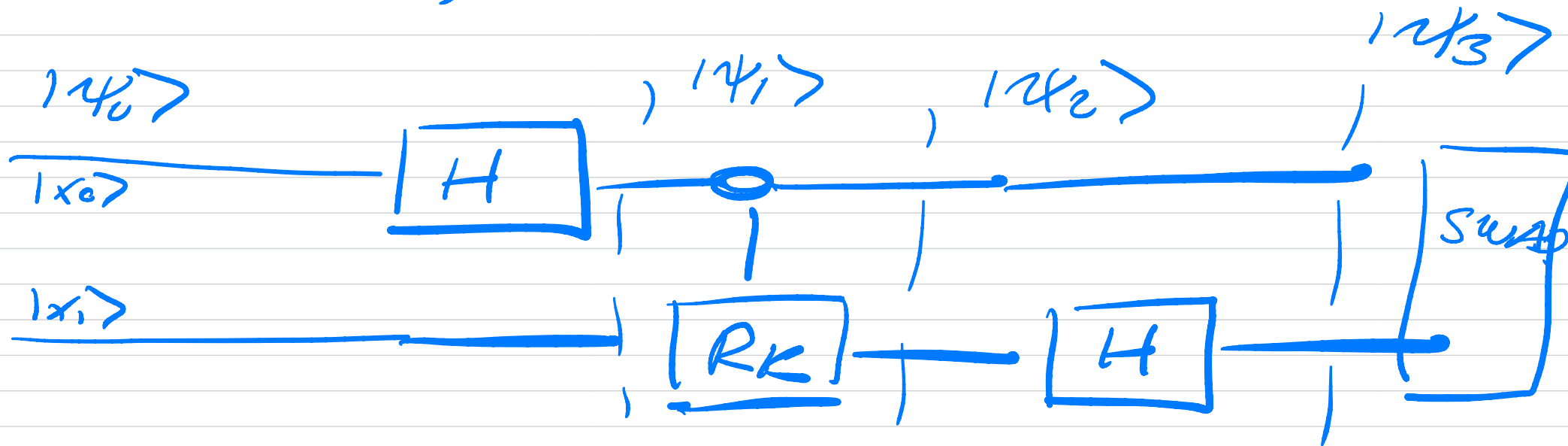
$$CS = \mathbb{1} \otimes R_k$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{\pi i / 2} \end{bmatrix}$$

$$\mathbb{I} \otimes H$$

$$QFT_2 = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & i & -i \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -i & i \end{bmatrix}$$

$$(I \otimes H) CS (H \otimes I)$$



$|x_0 x_1\rangle \rightarrow QFT$

$|x_0 x_1\rangle$
 $|y_0 y_1\rangle$

$$QFT_2 |3\rangle_2 =$$

$$\frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -i & -i \\ 1 & 1 & -1 & -1 \\ 1 & 1 & -i & i \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$= \frac{1}{2} (|00\rangle - i|01\rangle - |10\rangle + i|11\rangle)$$

$$QFT_2 |0\rangle_2 = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$= \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

QPE algorithm

unitary transformation

$$U \left(U U^\dagger = U^\dagger U = \mathbb{1} ; U = U^\dagger \right)$$

$$U = \begin{bmatrix} \vec{u}_0 & \vec{u}_1 & \vec{u}_2 & \dots & \vec{u}_{n-1} \\ 1 & 1 & 1 & & 1 \end{bmatrix}$$

$$u_i^\dagger u_j = \delta_{i,j}$$

has eigenvalue $\lambda_i = e^{2\pi i \phi_i}$

$$u \vec{u}_i = \lambda_i \vec{u}_i$$

$$\langle \vec{u}_i u^\dagger | u \vec{u}_i \rangle = |\lambda_i|^2$$

$$= |\lambda|^2$$

$$|\lambda| = 1 \Rightarrow \lambda = e^{2\pi i \phi}$$

$$\phi \in [0, 1]$$

Binary representation in terms of $-t-$ qubits

$$\phi = 0.\phi_0\phi_1\dots\phi_{t-1}$$

$$= \phi_0 \cdot \frac{1}{2^1} + \phi_1 \cdot \frac{1}{2^2} + \dots$$

$$\phi_{t-1} \cdot \frac{1}{2^t}$$

the t -bits represent
the precision we want
for ϕ .

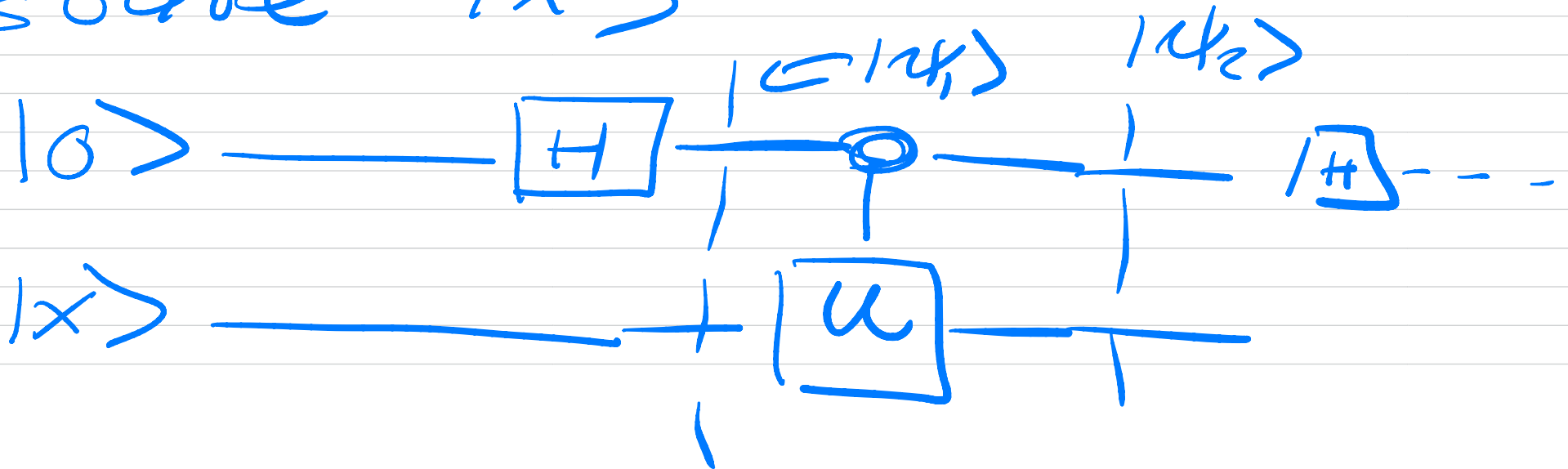
Example $t=1$

$$t=1 \quad \lambda = e^{2\pi i \cdot 0, \phi_0}$$

$$\phi_0 = \{0, 1\} \Rightarrow \phi = \{0, 1/2\}$$

Basic recipe

- initialize t -bits to $|0\rangle$
(First register)
- Second register contains
all the bits which are
needed to define the
state $|x\rangle$



$$|4_1\rangle = \frac{1}{\sqrt{2}} (|0\rangle \otimes |x\rangle + |1\rangle \otimes |x\rangle)$$

control - u

$$|4_2\rangle = \frac{1}{\sqrt{2}} (|0\rangle \otimes |x\rangle + e^{2\pi i \sigma, \phi_0} |1\rangle \otimes |x\rangle)$$

$$\otimes |x\rangle$$

$$= \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i \sigma, \phi_0} |1\rangle) \otimes |x\rangle$$

measuring either $|0\rangle$
or $|1\rangle$

$$H|\psi_2\rangle = |\psi_3\rangle =$$

$$\frac{1}{2} \left((1 + e^{2\pi i \sigma_z \phi_0}) |0\rangle + (1 - e^{2\pi i \sigma_z \phi_0}) |1\rangle \right) \otimes |x\rangle$$

ϕ_0 is either 0 or 1

$\phi_0 = 0$ then $|\psi_3\rangle = |0\rangle$

$\phi_0 = 1$ then $|\psi_3\rangle = |1\rangle$

if we measure with certainty
 $|0\rangle$ or $|1\rangle$, then $\phi = 0$
 or 0.5 , we know the
 eigenvalue since
 $\lambda = e^{2\pi i \phi}$

Example $n=2$

$$G, \phi_0 \phi_1 = \phi_0 \cdot \frac{1}{2} + \phi_1 \cdot \frac{1}{4}$$

$$\phi_0, \phi_1 = \{0, 1\} \Rightarrow \begin{array}{c} \bullet \bullet \bullet \bullet \\ \hline 0 \qquad \qquad \qquad 1 \end{array}$$

$$\phi = \{0, \frac{1}{2}, \frac{1}{4}, \frac{3}{4}\}$$